Motive · Machine Learning Auto-Marking System · Lecturer: NTIRANTA Jean Claude

Quiz report

Machine learning Quiz 1

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Year 4 IT

16 / 20

80% · Pass

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1. Section A: Multiple Choice (Q1–Q8) Circle the one correct answer. 1. Machine Learning is best defined as: No answer **0** / 1

- ☐ A. Writing explicit rules for every possible situation
- ☐ B. Enabling computers to learn from data and make predictions without being explicitly programmed ✓ correct
- ☐ C. Storing large amounts of data in a database
- ☐ D. Designing computer hardware for AI

2. 2. Which algorithm is used to predict a continuous numerical value such as house price? Correct **1** / 1

- ☐ A. Logistic Regression
- ☐ B. K-Means Clustering
- ☑ C. Linear Regression ✓ correct *(your answer)*
- ☐ D. Naive Bayes

3. 3. In machine learning, features are: Correct **1** / 1

- ☐ A. The output the model is trying to predict
- ☑ B. Independent variables used as input to the model ✓ correct *(your answer)*
- ☐ C. The errors made by the model
- ☐ D. The final trained model

4. 4. Which algorithm is an ensemble method that combines multiple decision trees to improve accuracy and reduce overfitting? Correct **1** / 1

- ☐ A. k-Nearest Neighbors
- ☐ B. Support Vector Machine

- [ ] C. Linear Regression
- [x] D. Random Forest ✓ correct *(your answer)*

5. 5. A model performs very well on training data but poorly on new, unseen data. This problem is called: No answer **0** / 1

- [ ] A. Underfitting
- [ ] B. Overfitting ✓ correct
- [ ] C. Data imbalance
- [ ] D. Feature scaling

6. 6. Which step of the ML process uses techniques such as Grid Search and Random Search? Correct **1** / 1

- [ ] A. Data collection
- [ ] B. Model deployment
- [x] C. Model tuning (hyperparameter optimization) ✓ correct *(your answer)*
- [ ] D. Data preprocessing

7. 7. Which of the following is an example of an unsupervised learning algorithm? Correct **1** / 1

- [ ] A. Decision Tree
- [x] B. K-Means Clustering ✓ correct *(your answer)*
- [ ] C. Linear Regression
- [ ] D. Logistic Regression

8. 8. Naive Bayes is based on Bayes' Theorem and assumes that: Correct **1** / 1

- [ ] A. All features are dependent on each other
- [ ] B. The data must always be labeled with three classes
- [x] C. Features are independent of each other ✓ correct *(your answer)*
- [ ] D. The model must have at least two hidden layers

9. Section B: True / False (Q9–Q14) Write TRUE or FALSE in the space provided. 9. Supervised learning trains a model using labeled data. Correct **1** / 1

Your answer: **True** · correct answer: **True**

10. Logistic Regression is mainly used to predict continuous values such as house prices. Correct **1** / 1

Your answer: **False** · correct answer: **False**

11. 11. Underfitting happens when a model is too simple and performs poorly on both training and test data. No answer **0** / 1

Your answer: **-** · correct answer: **True**

12. 12. In unsupervised learning, the training data always comes with labels. No answer **0** / 1

Your answer: **-** · correct answer: **False**

13. 13. k-Nearest Neighbors classifies a data point based on the majority class of its nearest neighbors Correct **1** / 1

Your answer: **True** · correct answer: **True**

14. 14. Reinforcement learning learns mainly from labeled input–output pairs provided by a teacher. Correct **1** / 1

Your answer: **False** · correct answer: **False**

15. Section C: Matching (Q15–Q20) Match each term (Q15–Q20) with its correct description (A–H). Each description is used once; two descriptions are extra. Correct **6** / 6

ITEM	YOUR MATCH	CORRECT MATCH
15. Support Vector Machine (SVM)	Finds the hyperplane that best separates different classes	Finds the hyperplane that best separates different classes
16. Principal Component Analysis (PCA)	A dimensionality reduction technique used to reduce the number of features	A dimensionality reduction technique used to reduce the number of features
17. Reinforcement Learning	Learns by interacting with an environment and receiving rewards or penalties	Learns by interacting with an environment and receiving rewards or penalties
18. Scikit-learn	A Python library for classical machine learning algorithms	A Python library for classical machine learning algorithms
19. SMOTE	A technique used to handle imbalanced datasets	A technique used to handle imbalanced datasets
20. Jupyter Notebook	An interactive environment for writing and testing machine learning code	An interactive environment for writing and testing machine learning code